# Integration: Concept

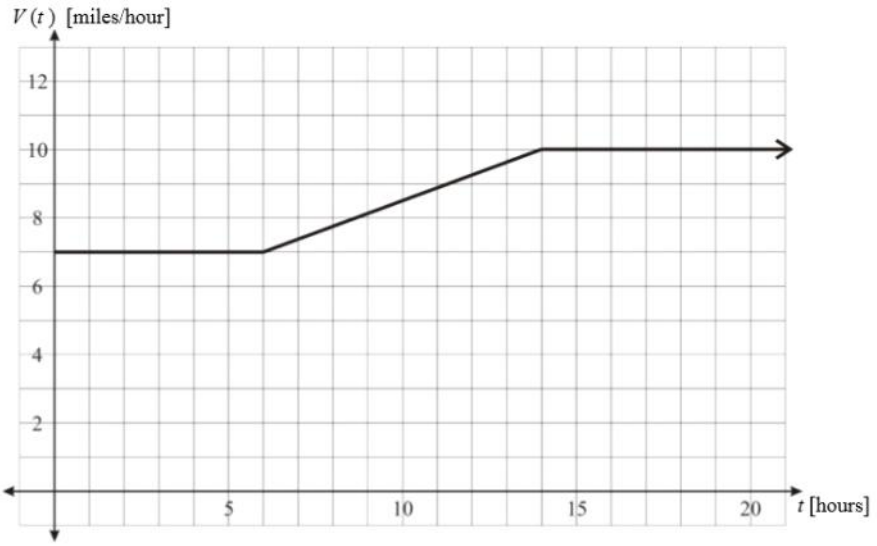
Monday, 26 October 2026 10:35 pm

***Calculus 1*** ***Handout 18***

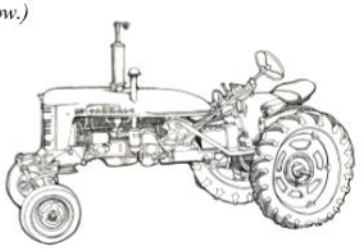
## Areas Bounded by Curves

Given a velocity function, we know that finding the distance traveled by the object, over an interval of time, involves finding the area bounded by the function's graph over that same interval.

Below is a hypothetical graph of the velocity of a tractor traveling on a straight road. Find the distance traveled by the tractor between $t = 6$ and $t = 18$.



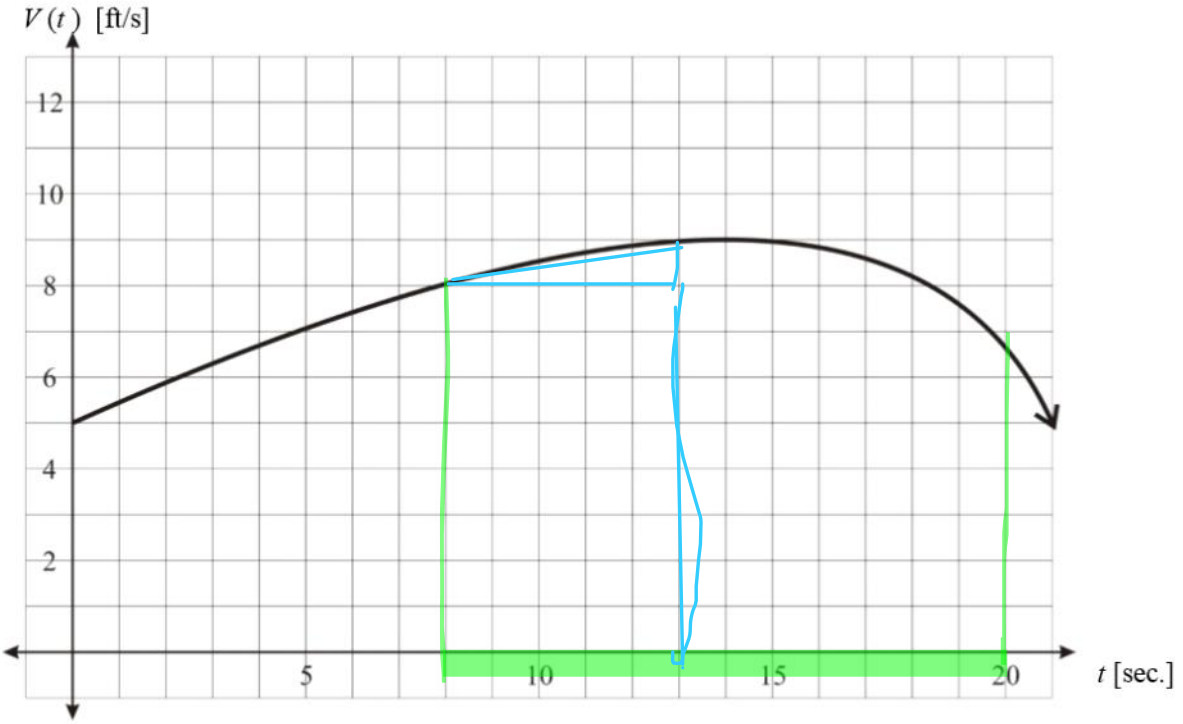

*(Document your method of approximating the area in the space below.)*



Area over the given interval = _____

Therefore, the distance traveled between the 6th and the 18th hour is _____ miles.

Next, let's consider the velocity function of a Victorian horse-carriage (graphed below). It is hard to envision being able to get an exact area, judging from the shape of the curve, but you can surely get an approximation. Use your creativity with various shapes and find the area under the curve over the interval [8, 20].




*(Document your method of approximating the area in the space below.)*

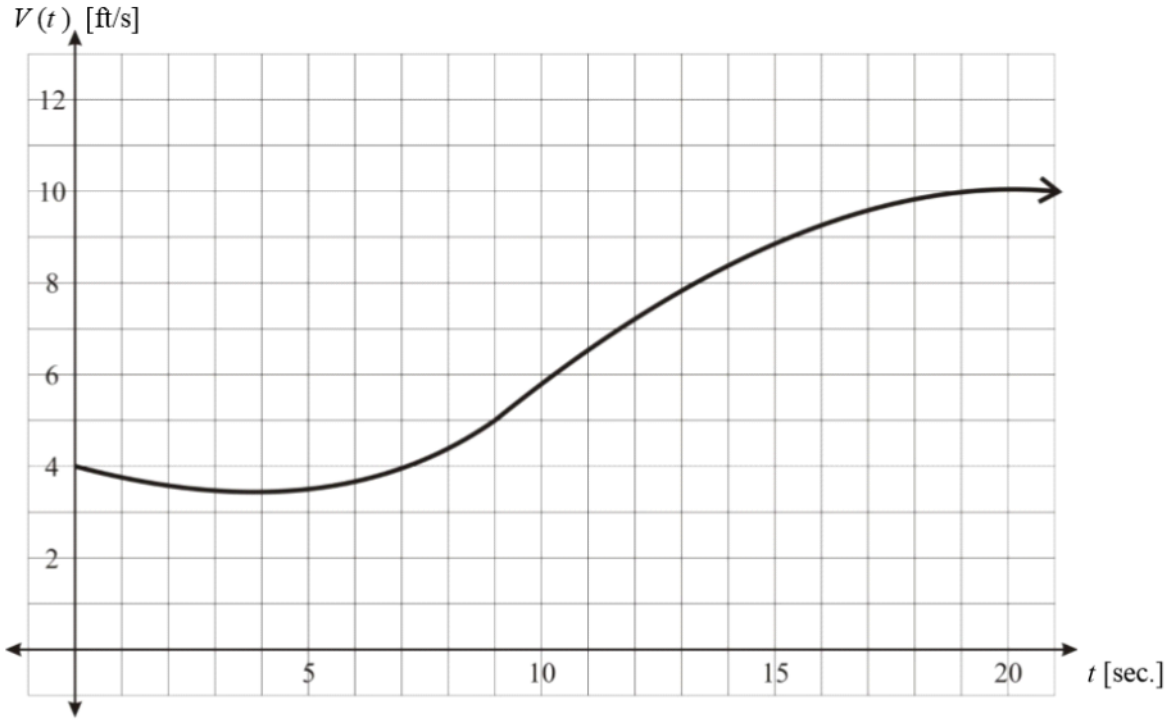


Area = _____

Therefore, the distance traveled between $t = 8$ and $t = 20$ is _____ feet.

*Now look up at the board as I point something out to you…..*

Graphed below is the velocity of Mr. Kerai running after the unidentified student who stole his special TI 89 calculator! Find an approximation for the distance traveled over the interval [4, 19], but this time restricting yourselves to rectangles of width 1 second as your primary shape.




Area = _____

Thus, the distance traveled between 4 and 19 seconds is _____ feet.

Now answer the following questions based on your observations of the work you have done so far.

- What is the correspondence between the velocity function's output and the height of any given rectangle you used?

- If you were given a formula or equation for the function, in addition to the graph, how would that help your calculations for the area of the rectangles.

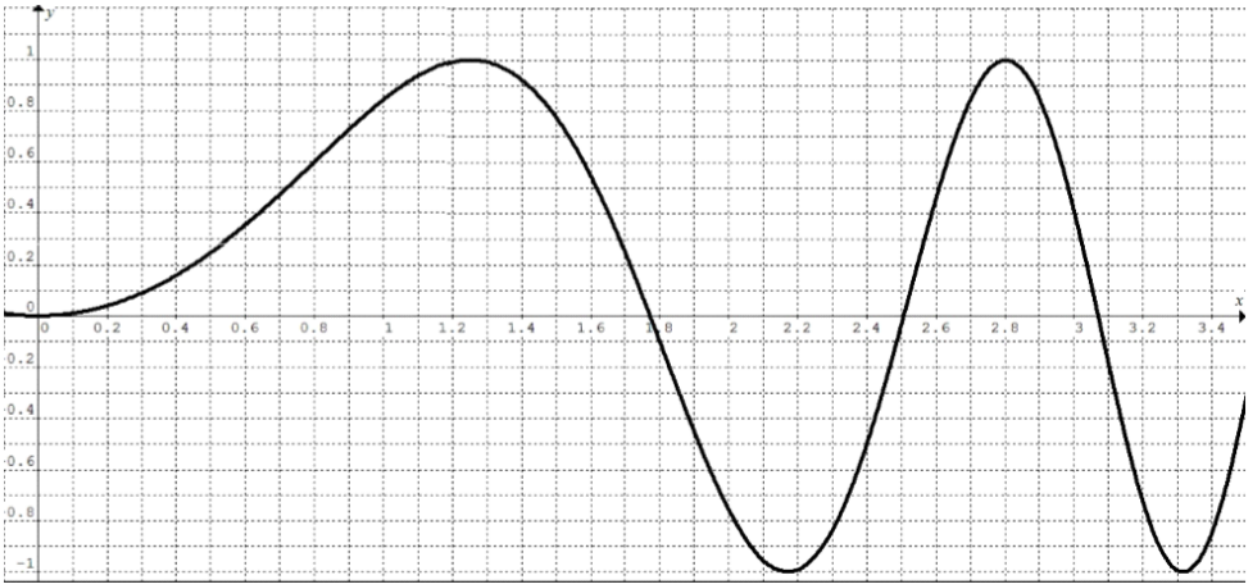
- If you were always restricted to using rectangles of uniform width to approximate areas bounded by curves, how could you improve the accuracy of your approximations?

*READING:*

*Read **pages 272 - 277** from the textbook and compare your reading to the work done today. Be sure to take notes on the reading.*

# More on Areas Bounded by Curves

The graph below is the graph of the function $f(x) = \sin(x^2)$.




Find the area bounded by the curve $f(x) = \sin(x^2)$ for $0.4 \le x \le 2.6$. (*Use rectangles of width 0.1 units.)*

Show all your work, i.e. write down all your individual calculations below. I have started you off for the first and second rectangles using left-hand sums, i.e. the top left corner of each rectangle touches the curve.

Area under the curve
for $0.4 \le x \le 2.6$ = $f(0.4) \times 0.1 + f(0.5) \times 0.1 +$